

22 DATA SECURITY

Computer security is the IT field dealing with security and protection of computing devices, computer systems and data. It covers many areas of protection against threats, e.g. unauthorized manipulation of computer devices, unauthorized data manipulation, malware, intrusion, attacks, software piracy, etc. It is based on key concepts: **confidentiality, integrity, availability, and authenticity of data**. It deals with security policies and methods such as general security measures and procedures, security technologies, security software, security solutions, copyright, etc. There are many types of computer security, for example, physical security, network security, cybersecurity, data security, information security, etc.

There is also a variety of **computer crimes** today committed by computer criminals, e.g. data theft, computer infections, malware, computer frauds like stolen credit card and financial data, social engineering, software piracy, cyber attacks and computer network attacks, hacking. A cracker is the hacker who breaks into other computer systems for fun, to steal information, or to propagate viruses. Crackers may intrude upon someone's privacy and cause serious damage. Malware refers to various forms of harmful software. To protect against malware we use antimalware software – security programs such as antispam (email filter), antivirus programs, antispyware, etc.

Social engineering	Tricking and manipulation of end-users to obtain confidential information, e.g. phishing, identity theft.
Software piracy	The illegal copying and distribution of copyrighted software.
Criminal hacking, crack	An intrusion to gain unauthorized access to networks systems and data.
Cyber attacks	Attacks on computer infrastructure to cause damage, obtain sensitive data, etc. Most common attacks on internet services are DoS attacks meaning Denial of Service to make a website or network resource unavailable. They block servers by flooding with requests to overload them.
Malware	Malware or malicious software is the general name for harmful software with malicious code which is created to damage or alter data.

Access control strategies

To protect data, computer or network from unauthorized access we use access control strategies like login, biometrics, authentication, etc.



Login is the process of identifying to enter a computer system or network. You usually type your username and password. There are several password requirements which a user has to comply with. Passwords must not be easy to guess. At the same time, they should not be too difficult for the user to remember. They should have a minimum of eight (or even ten) characters and a mixture of capital and small letters, numbers or symbols. Users should avoid using proper names, common words or personal data as passwords. People usually have more passwords, so it is better to use a password manager which can store and manage them.

Biometrics involves unique and measurable physical characteristics for identifying the user. It is also called biometric authentication or biometric access control: face recognition system, fingerprinting, iris or eye scanning, hand scanning, voice verification system, etc. Biometric devices: fingerprint reader, eye scanner, smart card reader, etc.

Selected security technologies

Firewall is a software and/or a hardware security device or system that monitors and controls the incoming and outgoing traffic. It allows limited access to an internal network from the Net, prevents unauthorized access to a computer or network. It can be a network device or personal firewall.

Proxy is a computer server which controls the traffic between the server and the client; mostly it sits between the Internet and the private network. It is used for a client application such as a web browser, email, etc.

Virtual private network (VPN) is a private computer network which is built over a public infrastructure and

allows remote users to access a local area network, e.g. a company's intranet. VPN ensures authentication, encryption and data integrity.

Monitoring includes tools and means to monitor a system, a network, an activity, etc. It is an effective way how to detect abnormal behaviour and activities.

Encryption, authentication methods, digital signature

Encryption is a process of saving and transmitting data in encoded form (encrypt = encode data, decrypt = decode data). There are a number of different encryption techniques.

Authentication is the verification of the identity of the user, entity or data integrity using different methods and usually requiring several security steps.

A **digital signature** guarantees that the information originated from the signer has not been changed since it was digitally signed. It can be used in email messages, macros, or electronic documents. Digital signatures are based on digital certificates, i.e. to create a digital signature, you need a signing certificate which proves identity.

Authorization is a process of granting rights to use a computer system (access rights).



Digital signature



Biometric access control

Types of unwanted programs, frauds and malicious software:

Type	Examples
Spam	Unsolicited email or message, junk mail, commercial advertising material, advertising messages.
Hoax	Fabricated, misleading news sent over the Net.
Phishing	Phishing (password harvesting fishing) is distributed via email or IM. It pretends to be a trusted message, while delivering a Trojan program or other malware to steal personal data. Users are taken to a false website, e.g. bank fraud to obtain banking information like passwords, credit card numbers, account details, etc.
Adware	Software that displays advertisements, e.g. pop-ups, some may include spyware.
Spyware	Software that collects information from your computer without permission.
Trojan	Software pretending to be a useful program that when activated may be destructive. Trojans do not self-replicate or reproduce by infecting other files as viruses.
Ransomware	The program can block up or lock your computer, encrypt all your data and ask for a ransom payment.
Self-copying programs: virus, worm	<i>Virus</i> – a program that spreads by attaching itself to files or other applications. It infects a computer system and reproduces itself. Opening attachments or files activates the virus. <i>Worm</i> – an autonomous program capable of creating its copies that does not need to be attached to a document or program. It replicates itself and spreads across a network, detects security holes in systems, etc.
Backdoor	Software (or hardware) which is hidden from the user can penetrate into the operating system without authentication.

VOCABULARY

access control /ˈæk.ses kən.trəʊl/	řízení přístupu
adware /ˈæd.weər/	reklamní software
antivirus program /ˌæn.tiˈvaɪə.rəs ˈprəʊ.græm/	antivirový program
attack /əˈtæk/	útok
authentication /ɔː.θen.tiˈkeɪʃn/	autentizace, ověření totožnosti
authorization /ɔː.θə.ər.aɪˈzeɪ.ʃən/	autorizace, oprávnění
availability /əˈveɪ.ləˈbɪl.ə.ti/	dostupnost
backdoor /ˈbæk.dɔːr/	zadní vrátka
biometric /ˌbaɪ.əʊˈmet.rɪk/	biometrický
confidentiality /ˌkɒn.fɪ.den.ʃiˈæl.ə.ti/	důvěrnost
cybercrime /ˈsaɪ.bə.kraɪm/	kybernetický zločin
cybersecurity /ˈsaɪbəsɪ.kjuərəti/	kybernetická bezpečnost
data theft /ˈdeɪ.tə ˌθeft/	krádež dat

Denial of Service /dɪˈnaɪ.əl əv ˈsɜː.vɪs/	odepření služby
fingerprinting /ˈfɪŋɡəprɪntɪŋ/	snímání otisků prstů
hacking /ˈhækɪŋ/	pronikání do systémů
hoax /həʊks/	klamná, poplašná zpráva
integrity /ɪnˈteg.rə.ti/	celistvost, neporušenost
intrusion /ɪnˈtruː.ʒən/	narušení, průnik
iris /ˈaɪ.rɪs/	oční duhovka
malware /ˈmæl.weər/	škodlivý software
obtain /əbˈteɪn/	získat, obdržet
password /ˈpɑːs.wɜːd/	heslo
phishing /ˈfɪʃɪŋ/	podvodné zprávy
proxy /ˈprɒk.si/	zástupný server
ransomware /ˈræn.səm.weər/	vyděračský software
social engineering /ˌsəʊ.ʃəl en.dʒɪˈniə.rɪŋ/	sociální inženýrství

software piracy	softwarové pirátství
/ˈsɒft.weər ˌpaɪ.rə.si/	
spyware	sledovací software
/ˈspaɪ.weər/	
threat	hrozba
/θret/	
Trojan (horse)	trojský kůň
/ˈtrɒj.ən.dʒən ˈhɔːs/	
unauthorized	neoprávněný
/ʌnˈɔː.θə.raɪzd/	
virus	virus
/ˈvaɪə.rəs/	
worm	červ
/wɜːm/	

► **TASKS** ◀

TASK 1

Security instructions and tips

Complete the following safety instructions and tips in the box with the words.

always	don't	never	mustn't
			give or hand over your password
			keep your password secret
			avoid giving your personal and financial information in social networks, in chat rooms, etc.
			use strong passwords, change your password after some time or frequently
			use out-of-date antivirus software
			run and update antivirus programs, use updates
			open email attachments from unknown senders
			make backup copies of all your important files regularly
			install a firewall with filtering level
			virus-check software before using it

TASK 2

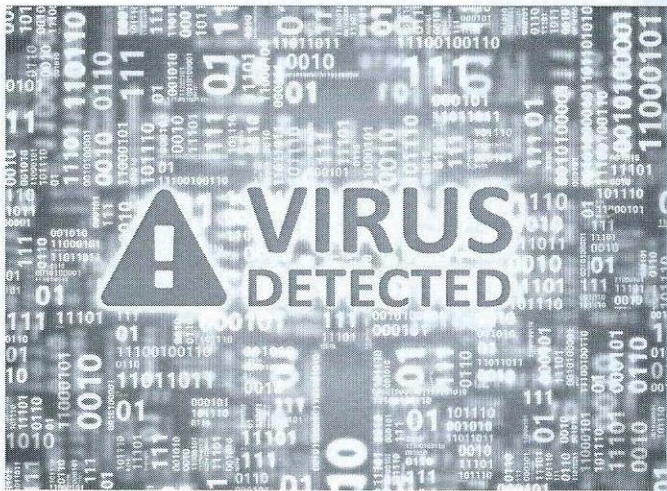
You and cybersecurity

Give some ideas showing:

- how to protect your personal data online and in social media
- how to use technologies safely
- how to create a strong password
- which security programs would you recommend

◀ **QUESTIONS** ▶

- 1 Which types of computer crimes and attacks do you know?
- 2 Explain these terms: encryption, authentication, malware, computer virus, Trojan.
- 3 Have you ever faced security issues? Which one(s)?
- 4 What do you use for computer security at home / at school?
- 5 How often do you update your computer, smartphone or tablet?



Security warning